

(c) Stars

(d) Moon

47th B.P.S.C. (Pre) 2005

Ans. (d)

Spring-tide occurs in the ocean on the day of Full Moon or New Moon, while on the 7th or 8th day after the Full Moon and the New Moon neap-tide occurs. Since, the Moon is closer to the Earth in comparison to other stars, so its gravitational force has a greater effect upon the Earth. Due to this effect, high sea waves are produced in oceans. Tide producing power of the Moon is 2.17 times more than that of the Sun.

The Oceanic Trenches

Oceanic Trenches are long narrow trenches that plunge to great depths of 5000 fathoms or 30000 feet. Trenches accounts for approximately 7% of the ocean's relief. Contrary to our expectations most of the deepest trenches are not located in the midst of the ocean; they are more often found close to the continents, particularly in the Pacific Ocean, where several deep trenches have been found.

***Mariana Trench** is the deepest trench in the world. It is located in the Pacific Ocean east of Mariana Islands. The Mariana Trench is a crescent-shaped trough. This trench was discovered in 1875 with the help of a survey ship called HMS Challenger. The Mariana Trench was also surveyed in 1951 by the British survey vessel, HMS Challenger II. The speculated depth of the Challenger Deep in Mariana Trench is 11,033 meters. The prominent trenches of the North Pacific Ocean are - Mariana Trench, Kuril Trench, Aleutian Trench and Japan Trench.

*The prominent trenches of Atlantic Ocean are - **Puerto Rico Trench** (N. Atlantic Ocean), **Romanche Trench** (Equatorial Atlantic Ocean) and South- **Sandwich Trench** (South Atlantic Ocean).

*The **Sunda trench** (Old name - Java trench) is the deepest trench in the Indian Ocean. Diamantina Trench is in the Indian Ocean. It is located about 1125 km southwest of the Perth Basin. It is also known as the Diamantina Fracture Zone.

1. Match List I with List II and select the correct answer using the code given below in the lists.

List- I

(Oceanic Trench)

A. Aleutian

B. Kermadec

C. Sunda

D. S. Sandwich

List- II

(Location)

1. Indian Ocean

2. North Pacific

3. South Pacific

4. South Atlantic Ocean

Code :

	A	B	C	D
(a)	2	4	1	3
(b)	2	3	1	4
(c)	1	3	2	4
(d)	1	4	2	3

I.A.S. (Pre) 2000

Ans. (b)

Trenches and deeps found in the oceans asked in the question are as follows –

Indian Ocean : Sunda /Java Trench and Diamantina Trench.

North Pacific Ocean : Mariana Trench, Kuril Trench, Aleutian Trench and Japan Trench.

South Pacific Ocean : Tonga Trench, Kermadec Trench, Atacama Trench.

North Atlantic Ocean : Puerto - Rico Trench.

South Atlantic Ocean : South Sandwich Trench.

Therefore, option (b) is the correct answer.

2. Which is the world's deepest trench?

- (a) Northern (b) Challenger
(c) Manhattan (d) Richards

U.P.P.C.S. (Pre) 1991

Ans. (b)

Challenger Trench is located in the Pacific Ocean, east of the Mariana Islands in the shape of a crescent. It has the deepest part of the Earth's oceans known as Challenger-Deep in Mariana Trench, named after the British exploration vessel HMS Challenger.

3. Which one of the following is the deepest oceanic trench?

- (a) Tonga (b) Mariana
(c) Puerto Rico (d) Izu-Bonin

U.P.P.C.S. (Mains) 2015

Ans. (b)

See the explanation of the above question.

4. In which one of the following oceans Diamantina Trench is situated ?

- (a) Pacific Ocean (b) Atlantic Ocean
(c) Indian Ocean (d) Arctic Ocean

I.A.S. (Pre) 2006

Ans. (c)

Diamantina Trench is in the Indian Ocean. It is located about 1125 km south-west of the Perth Basin. It is also known as Diamantina Fracture Zone.